

Home Credit Default Risk Prediction System

GitHub Repository

<https://github.com/Devansh1-tech/Home-Credit.git>

Project Overview

AI-powered loan default risk prediction system using the Home Credit Default Risk dataset, LightGBM, and SHAP Explainable AI.

Objectives

Predict default probability, reduce financial risk, assist loan officers, provide transparent explanations, and build a professional banking dashboard.

Dataset

Customer demographics, employment, income, bureau records, previous applications, credit card history, installments, POS cash balance. Target: 0=Repaid, 1=Default.

ML Pipeline

Data Cleaning -> Missing Value Imputation -> Feature Engineering -> Ordinal Encoding -> Feature Selection -> LightGBM -> Probability -> Risk Category -> SHAP Explainability.

Current Work

Completed preprocessing, feature engineering, model training, hyperparameter tuning, threshold optimization, SHAP notebooks, Streamlit prototype, prediction pipeline, helper utilities, and model serialization.

Models

best_lightgbm.pkl, mode_imputer.pkl, ordinal_encoder.pkl.

Risk Levels

Low (<30%): Approve. Medium (30-60%): Manual Review. High (>60%): Reject.

Explainable AI

SHAP Summary Plot, Waterfall Plot, Force Plot, Global Feature Importance, Local Explanation showing why a prediction was made.

Production Website

React + Tailwind CSS frontend, FastAPI backend, LightGBM inference API, responsive banking UI, customer prediction page, dashboards, model information page, explainability page, prediction history, PDF export, deployment on Vercel + Render.

Dashboard

Overview KPIs, risk distribution charts, prediction panel, customer profile, SHAP visualizations, model metrics, search and filtering.

Future Scope

Authentication, audit logs, role-based access, advanced analytics, cloud deployment, monitoring.